

CM0081 Formal Languages and Automata

§ 4.1 Proving Languages Not to Be Regular

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Preliminaries

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook (Hopcroft, Motwani and Ullman 2007).
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Properties of Regular Languages

- Proving languages not to be regular
- Closure properties
- Decision properties
- Equivalence and minimization of automata

The Pumping Lemma

Introduction

- Is $L_1 = \{ 0^m 1^n \mid m, n \geq 0 \}$ a regular language?

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- Is $L_4 = \{0^n 1^n \mid n \geq 1\}$ a regular language?

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Yes! $L_1 = L(0^* 1^*)$.

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Yes! $L_2 = L(0^+ 1^+)$.

- Is $L_3 = \{0^m 1^n \mid m \geq 2, n \geq 4\}$ a regular language?

Yes! $L_3 = L(000^* 11111^*)$.

- Is $L_4 = \{0^n 1^n \mid n \geq 1\}$ a regular language?

No! Informal proof (whiteboard).

The Pumping Lemma

Theorem 4.1 (Pumping Lemma for regular languages)

Let L be a regular language. Then there exists a positive integer n (which depends on L) such that for every string $w \in L$ such that $|w| \geq n$, we can break w into three strings, $w = xyz$, such that:

$$y \neq \varepsilon, \tag{1}$$

$$|xy| \leq n, \tag{2}$$

$$(\forall k \geq 0)(xy^kz \in L). \tag{3}$$

Formally,

$$(\exists n \in \mathbb{Z}^+)(\forall w \in L)(|w| \geq n \Rightarrow (\exists x)(\exists y)(\exists z)[w = xyz \wedge (1) \wedge (2) \wedge (3)]).$$

The Pumping Lemma

Proof

1. Suppose L is a regular language. Exist a DFA $A = (Q, \Sigma, \delta, q_0, F)$ with n states such that $L(A) = L$.

The Pumping Lemma

Proof

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2. Let $w = a_1 \cdots a_m \in L$, $m \geq n$ and $q_i = \hat{\delta}(q_0, a_1 \cdots a_i)$.

The Pumping Lemma

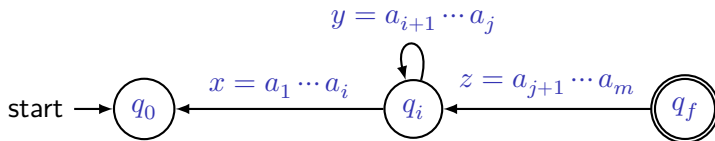
Proof

1. Suppose L is a regular language. Exist a DFA $A = (Q, \Sigma, \delta, q_0, F)$ with n states such that $L(A) = L$.
2. Let $w = a_1 \cdots a_m \in L$, $m \geq n$ and $q_i = \hat{\delta}(q_0, a_1 \cdots a_i)$.
3. By the pigeonhole principle, exists i and j , with $0 \leq i < j \leq n$ such that $q_i = q_j$.

The Pumping Lemma

Proof

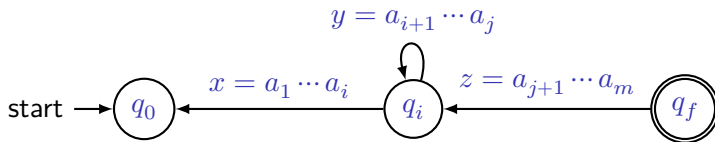
1. Suppose L is a regular language. Exist a DFA $A = (Q, \Sigma, \delta, q_0, F)$ with n states such that $L(A) = L$.
2. Let $w = a_1 \cdots a_m \in L$, $m \geq n$ and $q_i = \hat{\delta}(q_0, a_1 \cdots a_i)$.
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4. Let $w = xyz$ where



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Proof

1. Suppose L is a regular language. Exist a DFA $A = (Q, \Sigma, \delta, q_0, F)$ with n states such that $L(A) = L$.
2. Let $w = a_1 \cdots a_m \in L$, $m \geq n$ and $q_i = \hat{\delta}(q_0, a_1 \cdots a_i)$.
3. By the pigeonhole principle, exists i and j , with $0 \leq i < j \leq n$ such that $q_i = q_j$.
4. Let $w = xyz$ where



5. Then $(\forall k \geq 0)(xy^kz \in L)$. ■

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof schemata

Whiteboard.

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Exercise 4.1.2.e

Let $\Sigma = \{0, 1\}$ be an alphabet and let $L = \{ ww \mid w \in \Sigma^* \}$ be the so-called copy language. Prove that L is not regular.

(continued on next slide)

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof

1. Suppose L is regular.

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof

1. Suppose L is regular.
2. Let $n \in \mathbb{Z}^+$ be a constant (according to the Pumping Lemma).

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof

1. Suppose L is regular.
2. Let $n \in \mathbb{Z}^+$ be a constant (according to the Pumping Lemma).
3. Let $w = 0^n 10^n 1 \in L$ and $|w| \geq n$.

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof

1. Suppose L is regular.
2. Let $n \in \mathbb{Z}^+$ be a constant (according to the Pumping Lemma).
3. Let $w = 0^n 1 0^n \in L$ and $|w| \geq n$.
4. For the Pumping Lemma should exist x , y and z such that $w = xyz$, $|xy| \leq n$, $y \neq \varepsilon$ and $(\forall k \geq 0)(xy^kz \in L)$.

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2. Let $n \in \mathbb{Z}^+$ be a constant (according to the Pumping Lemma).
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4. For the Pumping Lemma should exist x , y and z such that $w = xyz$, $|xy| \leq n$, $y \neq \varepsilon$ and $(\forall k \geq 0)(xy^kz \in L)$.
5. For any x , y and z such that $w = xyz$, $|xy| \leq n$ and $y \neq \varepsilon$, we have that $y = 0^m$ with $0 < m \leq n$.

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3. Let $w = 0^n 1 0^n \in L$ and $|w| \geq n$.
4. For the Pumping Lemma should exist x , y and z such that $w = xyz$, $|xy| \leq n$, $y \neq \varepsilon$ and $(\forall k \geq 0)(xy^k z \in L)$.
5. For any x , y and z such that $w = xyz$, $|xy| \leq n$ and $y \neq \varepsilon$, we have that $y = 0^m$ with $0 < m \leq n$.
6. But, $xy^0 z \notin L$ which contradicts the Pumping Lemma.

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof

1. Suppose L is regular.
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3. Let $w = 0^n 1 0^n \in L$ and $|w| \geq n$.
4. For the Pumping Lemma should exist x , y and z such that $w = xyz$, $|xy| \leq n$, $y \neq \varepsilon$ and $(\forall k \geq 0)(xy^kz \in L)$.
5. For any x , y and z such that $w = xyz$, $|xy| \leq n$ and $y \neq \varepsilon$, we have that $y = 0^m$ with $0 < m \leq n$.
6. But, $xy^0z \notin L$ which contradicts the Pumping Lemma.
7. Therefore, L is not regular. ■

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Exercise 4.1.2.a

Let L be the language

$$L = \{ 0^n \mid n \text{ is a perfect square} \}.$$

Prove that L is not regular.

(continued on next slide)

Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof

1. Suppose L is regular.

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Application of the Pumping Lemma: Proving Languages Not to Be Regular

Proof

1. Suppose L is regular.
2. Let $n \in \mathbb{Z}^+$ be a constant (according to the Pumping Lemma).
3. Let $w = 0^{n^2} \in L$ and $|w| \geq n$.

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Proof

1. Suppose L is regular.
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1. Suppose L is regular.
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3. Let $w = 0^{n^2} \in L$ and $|w| \geq n$.
4. For the Pumping Lemma should exist x , y and z such that $w = xyz$, $|xy| \leq n$, $y \neq \varepsilon$ and $(\forall k \geq 0)(xy^kz \in L)$.
5. For any x , y and z such that $w = xyz$, $|xy| \leq n$ and $y \neq \varepsilon$, we have that $y = 0^m$ with $0 < m \leq n$ and $n^2 + 1 \leq |xyyz| \leq n^2 + n$.

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3. Let $w = 0^{n^2} \in L$ and $|w| \geq n$.
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5. For any x , y and z such that $w = xyz$, $|xy| \leq n$ and $y \neq \varepsilon$, we have that $y = 0^m$ with $0 < m \leq n$ and $n^2 + 1 \leq |xyz| \leq n^2 + n$.
6. Since the next perfect square after n^2 is $(n+1)^2 = n^2 + 2n + 1$, we know that $xyyz \notin L$ because $|xyyz|$ is strictly between the consecutive perfect squares n^2 and $(n+1)^2$.

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7. This a contradiction with the Pumping Lemma.

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6. Since the next perfect square after n^2 is $(n+1)^2 = n^2 + 2n + 1$, we know that $xyz \notin L$ because $|xyz|$ is strictly between the consecutive perfect squares n^2 and $(n+1)^2$.
7. This a contradiction with the Pumping Lemma.
8. Therefore, L is not regular. ■

Other Methods for Proving Languages Not to Be Regular

Observation

Frishberg and Gasarch (2018) show other methods and some open problems when proving that a language is not regular. The open problem 3.2 is related to the pumping lemma.

Open problem

“Find a non-regular language that cannot be proven non-regular using the pumping theorem and reductions, or show such a language does not exist.”

References



Daniel Frishberg and William Gasarch (2018). Open Problems Column. Different Ways to Prove a Language is Not Regular. SIGACT News 49.1, pp. 40–54. DOI: [10.1145/3197406.3197413](https://doi.org/10.1145/3197406.3197413) (cit. on p. 36).



John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).